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Robust Pricing–Hedging Duality and DeepMamba2 Integration for SPY Iron Condor Trading: A Technical Research Report

Abstract
This report is a rasical extension of its original presented by the author in 2017. It presents a comprehensive synthesis of robust pricing–hedging duality in continuous time, as formalized by Hou and Obłój (2015), with the CondorBrain™ DeepMamba2 neural architecture for institutional-grade SPY iron condor trading. We rigorously connect the pathwise superhedging framework, prediction sets, and martingale measures to the 58-dimensional feature space, 7 diffusion-based forward predictors, and 14 institutional trading rules of the CondorBrain system. The report derives new mathematical relationships linking prediction sets, martingale measures, and fuzzy logic-based position sizing, and details the integration of these concepts into the DeepMamba2 pipeline. Computational physics analogies, advanced feature engineering, and full Python code templates are provided for real-time M1 inference, robust risk control, and optimal entry/exit prediction. The resulting system is designed to maximize Sharpe ratio, minimize drawdown, and achieve superior net profit-to-drawdown (NP/DD) ratios, with production-grade engineering for tick data inference and institutional auditability.

Introduction

Motivation and Scope

Iron condor strategies on SPY are a cornerstone of institutional volatility trading, requiring robust pricing, precise risk management, and adaptive execution in the face of non-stationary, regime-dependent markets. Traditional model-specific approaches, while tractable, fail to account for model uncertainty and the full spectrum of market behaviors. The robust pricing–hedging duality framework, as developed by Hou and Obłój (2015), and extended by Bartl et al. (2020), provides a pathwise, model-agnostic foundation for pricing and hedging under uncertainty, leveraging prediction sets and martingale measures to interpolate between model-independent and model-specific regimes 1 2 3 4.

Recent advances in deep sequence modeling, particularly the Mamba-2 state-space architecture, have enabled the construction of neural trading systems capable of learning long-range dependencies, regime transitions, and complex market microstructure features at scale【scientific_spec.md】【Enhancing CondorBrain (Mamba-2) for Institutional-Grade SPY Options Trading.pdf】【research-Enhancing the Mamba 2 Neural Network for Advanced Trading and ROI Optimization.pdf】【Condor Brain Mathematical Ref.pdf】. The current CondorBrain DeepMamba2 system model creation process integrates a 58-dimensional institutional feature set, 7 diffusion-based forward predictors, and a suite of 14 institutional trading rules, with fuzzy logic-based position sizing and regime-aware risk controls.

This report aims to rigorously fuse the robust mathematical foundations of pathwise superhedging with the engineering and implementation details of the CondorBrain DeepMamba2

architecture, providing a blueprint for real-time, risk-aligned, and interpretable iron condor ETF trading on SPY m1, m5, and m15 multi-timeframe OHLCV – SPY Options Contract data in real-time.

Mathematical Framework

1. Robust Pricing–Hedging Duality and Pathwise Superhedging

1.1. Core Definitions
Let Ω denote the space of continuous price paths for the underlying and options, and let $\mathcal{X}$ be the set of European options available for static trading. The agent specifies a prediction set I ⊂ Ω, representing the set of paths deemed possible (beliefs), and a subset $\mathcal{P} \subset$ I for superhedging 2.

$$\Omega$$

Definitions: Ωis the space of continuous price paths for the underlying and options.

$$\mathcal{X}$$

Definitions: $\mathcal{X}$is the set of European options available for static trading.

$$I \subset \Omega$$

Definitions: $I$is the prediction set (belief set of paths deemed possible); Ωis the space of continuous price paths; ⊂denotes set inclusion.

$$\mathcal{P} \subset I$$

Definitions: $\mathcal{P}$is the subset of paths used for superhedging; $I$is the prediction set; ⊂denotes set inclusion.

A trading strategy consists of a static position X ∈ Lin_N($\mathcal{X}$) and a dynamic, progressively measurable process γ of bounded variation, satisfying admissibility constraints:

$$X \in \mathrm{Lin}_N(\mathcal{X})$$

Definitions: $X$is the static position; $\mathrm{Lin}_N(\mathcal{X})$is the set of linear combinations of at most Ninstruments in $\mathcal{X}$; $\mathcal{X}$is the set of European options; $N$is a fixed positive integer.

$$\gamma$$

Definitions: $\gamma$is a dynamic trading process (progressively measurable) of bounded variation.

$$\int_0^t \gamma_u\,(S)\,dS_u \le M\left(1+\sup_{0\le u\le t}|S_u\,|^p\right), \forall S \in \Omega, \forall t \in [0,T_n]$$

Definitions: $\int_0^t$ is the time integral from 0to t; $\gamma_u(S)$is the strategy evaluated at time $u$along path S; $dS_u$is the infinitesimal increment of the canonical price process at time u; $M$is a positive constant bounding admissible gains/losses; $\sup_{0\le u\le t}$is the supremum over $u \in [0,t]$; $|\,S_u\,|$is the absolute value (magnitude) of the process at time u; $p$is a fixed exponent controlling growth; $\forall$means "for all"; $S \in \Omega$means $S$is a continuous price path; $t \in [0,T_n]$means $t$lies in the interval from 0to terminal time T_n; $T_n$is a maturity/terminal horizon indexed by n.

A martingale measure P is a probability measure under which the canonical process S is a local martingale, and P(I) = 1. The set of calibrated market models $\boldsymbol{\mathcal{M}}$_{\mathcal{X}, \mathcal{P}, \mathcal{P}} consists of martingale measures that price all options in $\mathcal{X}$ correctly and are supported on $\mathcal{P}$.

$$P$$

Definitions: $P$is a probability measure (a candidate market model).

$$S$$

Definitions: $S$is the canonical price process (the coordinate mapping on the path space).

$$P(I) = 1$$

Definitions: $P(I)$is the probability (under P) of the prediction set I; $I$is the prediction set; 1means full probability mass (almost sure support on I).

$$\mathcal{M}_{\mathcal{X},\mathcal{P},\mathcal{P}}$$

Definitions: $\mathcal{M}_{\mathcal{X},\mathcal{P},\mathcal{P}}$is the set of calibrated martingale measures that (i) are consistent with prices of instruments in $\mathcal{X}$and (ii) are supported on $\mathcal{P}$; $\mathcal{X}$is the set of statically traded European options; $\mathcal{P}$is the superhedging set.

1.2. Superhedging and Duality
The minimal super-replicating cost for a contingent claim G on $\mathcal{P}$ is:

$$V_{X,P,P}(G) := \inf\left\{\mathrm{P(X)}:\exists(\mathrm{X},\gamma) \in A_X \mathrm{s.t.X(S)} + \int_0^{T_n} \gamma_u\,(\mathrm{S})\mathrm{d}S_u \geq \mathrm{G(S)},\forall \mathrm{S} \in \mathrm{P}\right\}$$

Definitions: $V_{\mathcal{X},\mathcal{P},\mathcal{P}}(G)$is the minimal superhedging (super-replicating) cost for claim G; inf denotes the infimum over feasible strategies; $\mathcal{P}(X)$is the (market) price/cost functional applied to the static position X; ∃means "there exists"; $(X,\gamma)$is a strategy pair (static X, dynamic γ); $\mathcal{A}_{\mathcal{X}}$is the admissible strategy set given static instruments $\mathcal{X}$; "s.t." means "such that"; $X(S)$is the payoff of the static position along path S; $\int_0^{T_n} \gamma_u\,(S)\,dS_u$is the gains process from dynamic trading up to T_n; ≥is inequality; $G(S)$is the claim payoff along path S; $\forall S \in \mathcal{P}$means for every path Sin the superhedging set $\mathcal{P}$; $T_n$is terminal time.

The robust price is:

$$P_{\mathcal{X},\mathcal{P},\mathcal{P}}(G) := \sup_{P \in \mathcal{M}_{\mathcal{X},\mathcal{P},\mathcal{P}}} \mathbb{E}_P[G(S)]$$

Definitions: $P_{\mathcal{X},\mathcal{P},\mathcal{P}}(G)$is the robust (worst-case) price of claim G; sup denotes supremum; $P \in \mathcal{M}_{\mathcal{X},\mathcal{P},\mathcal{P}}$means Pranges over calibrated martingale measures supported on $\mathcal{P}$; $\mathbb{E}_P[\cdot]$is expectation under measure P; $G(S)$is the payoff functional evaluated on canonical process S; $S$is the canonical price process.

Theorem (Pricing–Hedging Duality): Under suitable regularity, for bounded, uniformly continuous G:

$$V_{\mathcal{X},\mathcal{P},\mathcal{P}}(G) = P_{\mathcal{X},\mathcal{P},\mathcal{P}}(G)$$

Definitions: $V_{\mathcal{X},\mathcal{P},\mathcal{P}}(G)$is the minimal superhedging cost; $P_{\mathcal{X},\mathcal{P},\mathcal{P}}(G)$is the robust price; $G$is a bounded uniformly continuous contingent claim payoff; "=" denotes equality.

This result interpolates between model-independent (superhedging on all paths) and model-specific (unique martingale measure) settings, with the prediction set I quantifying the impact of beliefs and information 2 3 4.

$$I$$

Definitions: $I$is the prediction set encoding beliefs/information constraints on feasible paths.

1.3. Pathwise Superhedging on Prediction Sets
Bartl et al. (2020) extend this to pathwise superhedging: for a prediction set Ξ ⊆ C[0,T], the superhedging price for a claim ξ is:

$$\Xi \subseteq C[0,T]$$

Definitions: Ξis a prediction set of paths; ⊆means subset (possibly equal); $C[0,T]$is the space of continuous functions (paths) on the interval $[0, T]$; $T$is terminal time.

$$\Phi(\xi) := \inf\{\lambda \in \mathbb{R}: \exists (H^n), (G^n) \in \mathcal{H} \text{ s.t. } \lambda + (H^n \cdot S)_t + (G^n \cdot \mathbb{S})_t \geq 0 \text{ on } \Xi, \quad \forall n, t;\ \lambda + \liminf_{n\to\infty} ((H^n \cdot S)_T + (G^n \cdot \mathbb{S})_T) \geq \xi \text{ on } \Xi\}$$

Definitions: $\Phi(\xi)$is the pathwise superhedging price of claim ξ; inf is infimum; $\lambda$is an initial capital scalar; $\lambda \in \mathbb{R}$means $\lambda$is real-valued; ∃means "there exists"; $H^n$and $G^n$are sequences of trading strategies; $\mathcal{H}$is a specified admissible strategy class; "s.t." means "such that"; $(H^n \cdot S)_t$denotes the stochastic/pathwise integral (gains process) of strategy H^nagainst Sup to time t; $(G^n \cdot \mathbb{S})_t$denotes the corresponding gains against $\mathbb{S}$(typically a quadratic-variation-like or second-order process, depending on the framework) up to time t; $\mathbb{S}$is an auxiliary path-functional process used in the extension; ≥is inequality; "on Ξ" means pointwise for all paths in Ξ; $\forall n, t$means for all integers $n$and all times $t$in the trading horizon; $\liminf_{n\to\infty}$is the limit inferior; $T$is terminal time; $\xi$is the contingent claim payoff functional; Ξis the prediction set.

with dual representation:

$$\Phi(\xi) = \sup_{\mathbb{Q}\in\mathcal{M}(\Xi)} \mathbb{E}_{\mathbb{Q}}[\xi]$$

Definitions: $\Phi(\xi)$is the superhedging price; sup is supremum; $\mathbb{Q}$is a probability measure (candidate martingale measure); $\mathbb{Q} \in \mathcal{M}(\Xi)$means $\mathbb{Q}$is a martingale measure supported on Ξ; $\mathcal{M}(\Xi)$is the set of martingale measures supported on Ξ; $\mathbb{E}_{\mathbb{Q}}[\xi]$is the expectation of payoff ξunder $\mathbb{Q}$; $\xi$is the claim payoff.

where $\boldsymbol{\mathcal{M}}(\Xi)$ is the set of martingale measures supported on Ξ 1 2 3 4.

$$\mathcal{M}(\Xi)$$

Definitions: $\mathcal{M}(\Xi)$is the set of martingale measures supported on prediction set Ξ.

1.4. Martingale Optimal Transport and Model Uncertainty
The duality encompasses martingale optimal transport (MOT) problems, where the prediction set encodes marginal constraints (e.g., distributions of S_{T_1}, ..., S_{T_n}), and the robust price is the supremum over all martingale measures matching these marginals. This framework allows for model uncertainty and belief updating as new information or constraints are incorporated.

$$(S_{T_1}, \dots, S_{T_n})$$

Definitions: $S_{T_i}$is the canonical price process evaluated at time T_i; $T_1, \ldots, T_n$are specified maturities; the tuple denotes joint marginals/constraints.

1.5. Computational Considerations
• Discretization via Lebesgue partitions and piecewise constant path approximations.
• Construction of countable classes of approximating functions.
• Use of stochastic control and variational methods for constrained duality.
• Integration with Brownian motion-based models and Lagrange multipliers for calibration.

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2. Prediction Sets, Martingale Measures, and Fuzzy Logic Sizing

2.1. Prediction Sets in Practice
In the context of SPY iron condor trading, the prediction set I can be constructed using:
• Historical volatility regimes (e.g., IVR > 50 for high-volatility).
• Exclusion of paths violating liquidity or microstructure constraints.
• Incorporation of institutional signals (e.g., block trades, volume spikes, regime shifts).
This enables the system to quantify the impact of assumptions and dynamically update beliefs as new data arrives.

$$\text{IVR} > 50$$

Definitions: IVRis implied volatility rank; 50is the threshold level; “>” means greater than.

2.2. Martingale Measures and Calibrated Models
The set of martingale measures $\boldsymbol{\mathcal{M}}$^I is restricted to those supported on I, and further to those calibrating to observed option prices and Greeks. In the CondorBrain system, this calibration is performed via:
• Real-time alignment of spot and options chain data (lag-aware, as-of joins).
• Feature engineering to ensure all model inputs are causally valid and free from lookahead bias 【CondorBrain™ – CondorIntelligence™ Input Data and Model Creation Process.pdf】【walkthrough.md】.

$$\mathcal{M}^I$$

Definitions: $\mathcal{M}^I$is the set of martingale measures supported on prediction set I; $I$is the prediction set.

2.3. Fuzzy Logic-Based Position Sizing
Fuzzy logic provides a measure-theoretic framework for mapping prediction set membership and regime probabilities to position sizes. The fuzzy logic engine:

• Aggregates multi-factor indicator signals (RSI, ADX, IVR, regime consensus) into a confidence score μ_{fuzzy} ∈ [0,1].
• Combines this with neural confidence φ_{mamba} and risk factors to compute the final trade size:

$$\mu_{\text{fuzzy}} \in [0,1]$$

Definitions: $\mu_{\text{fuzzy}}$is the fuzzy confidence score (membership/confidence); $[0, 1]$is the unit interval; "∈" means "is an element of".

$$S_{\text{trade}} = S_{\max} \cdot (w_1 \cdot \mu_{\text{fuzzy}} + w_2 \cdot \phi_{\text{mamba}}) \cdot \text{RiskFactor}$$

Definitions: $S_{\text{trade}}$is the final trade size; $S_{\max}$is the maximum allowable position size; $w_1$is a weight on fuzzy confidence; $\mu_{\text{fuzzy}}$is the fuzzy confidence score; $w_2$is a weight on neural confidence; $\phi_{\text{mamba}}$is the neural confidence output from the DeepMamba2 model; RiskFactoris a multiplicative risk control term (e.g., based on drawdown, max loss, regime constraints); "·" denotes multiplication; "+" denotes addition; "=" denotes equality.

• Ensures that position sizing is risk-aligned, interpretable, and responsive to both model predictions and rule-based constraints【Institutional-Grade Trading Rules for SPY Options Model Integration.pdf】【CondorBrain™ – CondorIntelligence™ Input Data and Model Creation Process.pdf】【Enhancing CondorBrain (Mamba-2) for Institutional-Grade SPY Options Trading.pdf】.

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3. Computational Physics Analogies
   The pathwise superhedging framework and DeepMamba2 architecture can be interpreted through the lens of computational physics:
   • State-Space Models (SSMs): Analogous to dynamical systems, where the latent state h(t) evolves according to ODEs:

$$\dot{h}(t) = \mathbf{A}h(t) + \mathbf{B}x(t)$$

Definitions: $\dot{h}(t)$is the time derivative of latent state $h(t)$; $h(t)$is the latent state vector at time t; **A**is the state transition matrix/operator; **B**is the input matrix/operator; $x(t)$is the input feature vector at time t; "=" is equality; "+" is addition.

$$y(t) = \mathbf{C}h(t) + \mathbf{D}x(t)$$

Definitions: $y(t)$is the output vector at time t; **C**is the readout matrix/operator from latent state to output; $h(t)$is the latent state; **D**is the direct-feedthrough matrix/operator from inputs to outputs; $x(t)$is the input vector; "=" is equality; "+" is addition.

• Selective Scan Mechanism: Implements an associative scan (parallel prefix sum) over the sequence, enabling linear-time inference and dynamic memory management, akin to propagating physical states through a discretized medium【scientific_spec.md】【Novel Mathematical Enhancements for CondorBrain Mamba.pdf】.
• Manifold and Topological Features: Embedding high-dimensional market states onto nonlinear manifolds (e.g., torus, sphere) and analyzing curvature, persistent homology, and chaos signatures mirrors techniques in statistical mechanics and topology for characterizing phase transitions and regime shifts【Dynamic Market Indicators - Math and Pythonic integration for CondorIntelligence.pdf】.

Model Integration: DeepMamba2 and CondorBrain System

1. Feature Engineering and Data Pipeline

1.1. 58-Dimensional Institutional Feature Set
The CondorBrain system ingests a 58-dimensional feature vector per time step, including:
• Market and Options State: OHLCV, strike, call/put, delta, gamma, vega, theta, IV, IVR, spread ratio, time-to-expiry, SMA, PSAR, target spot, max drawdown.
• Dynamic Features: Log returns, EWMA volatility, normalized returns, ATR%, curvature proxy, volatility energy, dynamic RSI, adaptive ADX, PSAR, dynamic Bollinger bands, stochastic oscillator, consolidation and breakout scores.
• Primitive Features: Bandwidth, percentile, expansion rate, volume ratio, friction, execution gates, gap risk, IV confidence, multi-timeframe consensus, MACD, PSAR trend, chaos membership, fuzzy reversion.
All features are precomputed and aligned using a lag-aware, as-of join pipeline, ensuring strict causality and eliminating lookahead bias【CondorBrain™ – CondorIntelligence™ Input Data and Model Creation Process.pdf】【walkthrough.md】.

1.2. Diffusion-Based Forward Predictors
Seven diffusion-based predictors model forward returns, volatility, high-low envelopes, regime shifts, liquidity shocks, curvature energy, and chaos membership, providing probabilistic forecasts for future price and volatility trajectories【CondorBrain™ – CondorIntelligence™ Input Data and Model Creation Process.pdf】.

1.3. Institutional Trading Rules
Fourteen institutional-grade trading rules are implemented as primitives, including trend continuation, volatility-adaptive breakout, fuzzy reversion, multi-timeframe consensus, regime scoring, curvature and liquidity-aware reversion, spread friction gating, volatility expansion, chaos override, gap risk, position size scaling, and composite entry logic. Each rule is

mathematically defined and integrated as a feature or constraint in the model pipeline 【Institutional-Grade Trading Rules for SPY Options Model Integration.pdf】 【CondorBrain™ – CondorIntelligence™ Input Data and Model Creation Process.pdf】.

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2. DeepMamba2 Architecture

2.1. Core Structure

• Input Embedding: Projects the 58-dimensional feature vector into a high-dimensional latent space (typically 512–1024 dimensions) using layer normalization and linear expansion.
• Mamba-2 Backbone: A stack of 24–32 selective-scan SSM layers, each implementing input-adaptive state transitions and gating, enabling the model to learn long-range dependencies and regime transitions efficiently.
• Volatility-Gated Attention: Sparse attention layers are inserted at key depths (e.g., layers 8, 16, 24), activated only during high-volatility regimes, preserving linear-time scaling while enabling global context when needed【Enhancing CondorBrain (Mamba-2) for Institutional-Grade SPY Options Trading.pdf】【Novel Mathematical Enhancements for CondorBrain Mamba.pdf】【Integrated Imp.pdf】.
• Regime-Routed Mixture-of-Experts (MoE): The final hidden state is routed through a softmax gating network to a set of regime-specialized expert heads (e.g., low, normal, high volatility), each outputting iron condor policy parameters.
• Diffusion Forecaster: Generates multi-horizon price and volatility forecasts, informing strike selection and risk management.
• Fuzzy Logic Fusion: Aggregates neural outputs, technical indicators, and rule-based signals into a final position size and trade intent, using membership functions and Q-table policy mapping.

2.2. Output Heads

• Iron Condor Policy Head: Predicts optimal call/put offsets, wing width, DTE, probability of profit, expected ROI, max loss, and confidence.
• Regime Detector: Classifies the current volatility regime (low, normal, high).
• Horizon Forecaster: Outputs multi-day price trajectories and volatility envelopes.
• Fuzzy Sizing Head: Computes the final position size as a function of neural and fuzzy logic confidence.

2.3. Loss Function and Training Objectives

The composite risk-aligned loss integrates:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{pred}} + \lambda_2 \mathcal{L}_{\text{dir}} + \lambda_3 \mathcal{L}_{\text{Sharpe}} + \lambda_4 \mathcal{L}_{\text{DD}} + \lambda_5 \mathcal{L}_{\text{turn}}$$

Definitions: $\mathcal{L}_{\text{total}}$is the total composite loss; $\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$are nonnegative scalar weights; $\mathcal{L}_{\text{pred}}$is the predictive-fidelity loss term; $\mathcal{L}_{\text{dir}}$is the directional loss term; $\mathcal{L}_{\text{Sharpe}}$is a negative

Sharpe proxy loss term; $\mathcal{L}_{\mathrm{DD}}$is the drawdown penalty term; $\mathcal{L}_{\mathrm{turn}}$is the turnover penalty term; "=" is equality; "+" is addition.

• $\mathcal{L}_{\mathrm{pred}}$: Huber loss for predictive fidelity.
• $\mathcal{L}_{\mathrm{dir}}$: Directional loss (e.g., MADL).
• $\mathcal{L}_{\mathrm{Sharpe}}$: Negative Sharpe ratio proxy.
• $\mathcal{L}_{\mathrm{DD}}$: Soft drawdown penalty (log-sum-exp of drawdowns).
• $\mathcal{L}_{\mathrm{turn}}$: Turnover penalty for position changes.
This loss directly aligns training with trading objectives: maximizing risk-adjusted returns, minimizing drawdown, and controlling slippage【Enhancing CondorBrain (Mamba-2) for Institutional-Grade SPY Options Trading.pdf】【Novel Mathematical Enhancements for CondorBrain Mamba.pdf】【Integrated Imp.pdf】【Condor Brain Mathematical Ref.pdf】.

2.4. Computational Engineering
• BF16/TF32 Mixed Precision: Enables high-speed, high-precision training and inference on A100/H100 GPUs.
• CUDA Graphs: Sub-second inference via static CUDA graph capture.
• Gradient Accumulation and Checkpointing: Memory-efficient training with large effective batch sizes.
• Memory Fragmentation Control: Environment variables (e.g., PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True) prevent OOM errors.
• LazySequenceDataset: Zero-copy, memory-efficient data loading for 10M+ row datasets.

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3. Mathematical Derivations: Linking Prediction Sets, Martingale Measures, and Fuzzy Sizing

3.1. Prediction Set–Martingale Measure Correspondence
Given a prediction set I, the set of admissible martingale measures $\boldsymbol{\mathcal{M}}$^I is:

$$\mathcal{M}^{I} = \{\, P \in \mathcal{M} : P(I) = 1 \,\}$$

Definitions: $\mathcal{M}^{I}$is the set of martingale measures supported on prediction set I; $\{\ \cdot\ \}$denotes a set; $P$is a probability measure; $P \in \mathcal{M}$means Pbelongs to the baseline set of martingale measures $\mathcal{M}$; $P(I) = 1$means Passigns full probability to set I; ":" means "such that"; "=" is equality.

For any claim G, the robust price is:

$$P_{\mathcal{X},\mathcal{P},I}(G) = \sup_{P \in \mathcal{M}_{\mathcal{X},\mathcal{P},I}} \mathbb{E}_P[G(S)]$$

Definitions: $P_{\mathcal{X},\mathcal{P},I}(G)$is the robust price of claim Gunder static instruments $\mathcal{X}$, superhedging set $\mathcal{P}$, and prediction set I; sup is supremum; $P$is a probability measure; $\mathcal{M}_{\mathcal{X},\mathcal{P},I}$is the set of

calibrated martingale measures supported on $I$and consistent with $\mathcal{X}$and $\mathcal{P}$; $\mathbb{E}_P[\cdot]$is expectation under P; $G(S)$is payoff evaluated on canonical process S; $S$is the canonical price process; "=" is equality.

where $\mathcal{M}_{\mathcal{X},\mathcal{P},I}$is the set (collection) of **calibrated martingale probability measures** used as admissible market models; $\mathcal{X}$denotes the set of statically tradable American ETF options (the calibration instrument set); $\mathcal{P}$denotes the superhedging path set (the set of paths on which inequalities must hold); Idenotes the prediction set (the belief/information constraint set of paths); the subscript "$_{\mathcal{X},\mathcal{P},I}$" indicates that the measures in the set are constrained by $\mathcal{X}$, $\mathcal{P}$, and I.

calibrates to observed prices and is supported on I.

$$\mathcal{M}_{\mathcal{X},\mathcal{P},I}$$

Definitions: $\mathcal{M}_{\mathcal{X},\mathcal{P},I}$is the set of martingale measures calibrated to $\mathcal{X}$, supported on I, and consistent with the superhedging set $\mathcal{P}$.

3.2. Fuzzy Logic as a Measure on Prediction Sets
Let μ_{fuzzy}(x) denote the fuzzy membership of state x in the "safe to trade" set. The fuzzy position size is:

$$S_{\text{trade}}(x) = S_{\max} \cdot \mu_{\text{fuzzy}}(x) \cdot \phi_{\text{mamba}}(x) \cdot \text{RiskFactor}(x)$$

Definitions: $S_{\text{trade}}(x)$is the position size as a function of state x; $S_{\max}$is maximum allowed size; $\mu_{\text{fuzzy}}(x)$is fuzzy membership/confidence for state x; $\phi_{\text{mamba}}(x)$is neural confidence for state x; $\text{RiskFactor}(x)$is a risk control multiplier for state x; "·" is multiplication; "=" is equality.

where φ_{mamba}(x) is the neural confidence, and RiskFactor(x) encodes risk controls (e.g., max loss, drawdown).
This can be interpreted as integrating over the belief-weighted prediction set:

$$\mathbb{E}_{\mu_{\text{fuzzy}}}[S_{\text{trade}}] = \int_I S_{\text{trade}}(x)\, d\mu_{\text{fuzzy}}(x)$$

Definitions: $\mathbb{E}_{\mu_{\text{fuzzy}}}[S_{\text{trade}}]$is an expectation of trade size under the fuzzy-weighted measure; $\int_I$ is an integral over the prediction set I; $S_{\text{trade}}(x)$is trade size at state x; $d\mu_{\text{fuzzy}}(x)$is the differential of the fuzzy measure over states; $I$is the prediction set; "=" is equality.

3.3. Composite Loss and Risk Alignment

The composite loss function ensures that the model not only predicts accurately but also optimizes for risk-adjusted returns and drawdown minimization. The soft drawdown penalty is:

$$\mathcal{L}_{\mathrm{DD}} = \tau \log \left(\sum_t \exp \left(\frac{\mathrm{DD}_t}{\tau} \right) \right)$$

Definitions: $\mathcal{L}_{\mathrm{DD}}$ is the drawdown penalty term; τ is a positive smoothing temperature parameter; $\log(\cdot)$ is the natural logarithm; $\sum_t$ sums over time index t; $\exp(\cdot)$ is the exponential function; DD_t is drawdown at time t; "/" denotes division; "=" is equality.

where DD_t is the drawdown at time t, and τ is a smoothing parameter.

$$\mathrm{DD}_t$$

Definitions: DD_t is drawdown at time t (peak-to-trough loss measured at t).

$$\tau$$

Definitions: τ is a positive smoothing parameter controlling soft-max behavior in the drawdown penalty.

The Sharpe proxy is:

$$\mathcal{L}_{\mathrm{Sharpe}} = -\frac{\mu}{\sigma + \epsilon}$$

Definitions: $\mathcal{L}_{\mathrm{Sharpe}}$ is the negative Sharpe proxy loss; μ is the mean of returns over an evaluation window; σ is the standard deviation of returns over the same window; ϵ is a small positive constant for numerical stability; the leading "–" indicates negation; "/" is division; "+" is addition; "=" is equality.

where μ and σ are the mean and standard deviation of returns.

$$\mu$$

Definitions: μ is the mean (average) return over a chosen window.

$$\sigma$$

Definitions: $\sigma$is the standard deviation (volatility) of returns over a chosen window.

$$\epsilon$$

Definitions: $\epsilon$is a small positive numerical-stability constant (e.g., 10^{-9}).

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4. Python Implementation: DeepMamba2 Pipeline
   Below is a template for integrating the robust pricing–hedging duality and fuzzy logic sizing into the DeepMamba2 pipeline.

```python
import torch
import torch.nn as nn

class CompositeCondorLoss(nn.Module):
    def __init__(self, lambdas=(1.0, 0.5, 0.1, 0.1, 0.05)):
        super().__init__()
        self.lambdas = lambdas
        self.huber = nn.HuberLoss(delta=1.0)

    def forward(self, y_pred, y_true, returns, weights, last_weights):
        # 1. Predictive Fidelity (Huber)
        error = y_true - y_pred
        l_pred = torch.where(torch.abs(error) <= 1.0,
                             0.5 * error ** 2,
                             1.0 * (torch.abs(error) - 0.5)).mean()
        # 2. Sharpe Proxy
        mu, sigma = returns.mean(), returns.std()
        l_sharpe = - (mu / (sigma + 1e-9))
        # 3. Soft Drawdown Penalty
        cum_ret = returns.cumsum(dim=-1)
        peak = torch.cummax(cum_ret, dim=-1).values
        dd = peak - cum_ret
        l_dd = torch.log(torch.exp(dd / 0.02).sum(dim=-1) + 1e-9).mean()
        # 4. Turnover Penalty
        l_turn = (weights - last_weights).abs().mean()
        return (l_pred +
                self.lambdas[1] * l_sharpe +
                self.lambdas[2] * l_dd +
                self.lambdas[3] * l_turn)

```

Integration Steps:

1. Feature Engineering: Precompute all 58 features, including dynamic, primitive, and topological indicators, and align spot/options data using lag-aware as-of joins.
2. Model Construction: Instantiate the DeepMamba2 model with volatility-gated attention and regime-routed MoE heads.
3. Loss Function: Replace pure Huber loss with the composite risk-aligned loss.

4. Fuzzy Logic Sizing: Implement the fuzzy logic engine as a post-processing layer, aggregating neural and indicator signals into a final position size.
5. Risk Controls: Enforce drawdown, turnover, and regime constraints at both the model and execution layers.
6. Production Engineering: Enable BF16/TF32 mixed precision, CUDA graphs, and memory optimizations for sub-second inference.

Computational Physics and Topological Feature Engineering

1. Manifold Geometry and Curvature
   • Curvature Proxy: Compute the second difference of log returns, normalized by local scale, as a proxy for local market curvature:

$$\kappa_t = \frac{\mathrm{EMA}_{64}(\ddot{r}_t)}{\sigma_t + \epsilon}$$

Definitions: $\kappa_t$is the curvature proxy at time t; $\mathrm{EMA}_{64}(\cdot)$is an exponential moving average with span/window parameter 64; $\ddot{r}_t$is the second difference (discrete second derivative) of log returns at time t; $\sigma_t$is EWMA volatility at time t; $\epsilon$is a small positive constant for numerical stability; "/" is division; "+" is addition; "=" is equality.

where r̈_t = r_t − 2r_{t−1} + r_{t−2}, and σ_t is EWMA volatility.

$$\ddot{r}_t = r_t - 2r_{t-1} + r_{t-2}$$

Definitions: $\ddot{r}_t$is the discrete second difference of log returns; $r_t$is the log return at time t; $r_{t-1}$is the log return at time $t-1$; $r_{t-2}$is the log return at time $t-2$; constants 2and signs "−" and "+" are standard arithmetic; "=" is equality.

$$\sigma_t$$

Definitions: $\sigma_t$is the EWMA volatility at time t.

• Volatility Energy: Define volatility energy as:

$$E_t = \ln\ (1+\alpha\ |\ \kappa_t\ |)$$

Definitions: $E_t$is volatility energy at time t; ln (·)is the natural logarithm; 1is the constant one; $\alpha$is a nonnegative scaling parameter; | κ_t |is the absolute value (magnitude) of curvature proxy κ_t; $\kappa_t$is curvature proxy at time t; "+" is addition; "·" is multiplication; "=" is equality.

High energy signals regime transitions and volatility spikes.

2. Persistent Homology and Topological Regimes
   • Persistent Homology Signature: Embed recent price or IV sequences using Takens delay maps, compute persistence lifetimes of 1D homology classes, and sum the largest J lifetimes as the regime signature Π_t.
   • Interpretation: High, stable Π_t indicates consolidation; sharp drops signal breakouts or regime shifts.

J

Definitions: $J$is the number of largest persistence lifetimes selected/summed to form the signature.

Π_t

Definitions: $\Pi_t$is the persistent-homology-based regime signature at time t(sum of top Jlifetimes).

3. Pythonic Implementation

```
def curvature_proxy_from_returns(r, span=64):
    dr = r.diff()
    d2r = dr.diff()
    scale = dr.abs().ewm(span=span, adjust=False).mean() + 1e-12
    kappa = d2r / scale
    return kappa.ewm(span=max(8, span // 4), adjust=False).mean()

def volatility_energy_from_curvature(kappa, alpha=1.0):
    u = kappa.abs().astype(float)
    return np.log1p(alpha * u)

```

Real-Time M1 Inference and Production Engineering

1. Data Pipeline and Causality
   • Lag-Aware Alignment: All spot and options features are aligned using as-of joins, with IV confidence decaying exponentially with lag, ensuring no lookahead bias.
   • Multi-Timeframe Consensus: Signals are aggregated across 1m, 5m, and 15m bars for regime confirmation.
2. GPU Optimization
   • BF16/TF32 Mixed Precision: All model weights and activations are stored in BF16 for maximum throughput on A100/H100 GPUs.
   • CUDA Graphs: Inference is captured in static CUDA graphs for sub-millisecond latency.
   • Gradient Accumulation and Checkpointing: Enables effective batch sizes of 256+ without OOM errors.
3. Inference Engine Template

```
class CondorBrainEngine:
    def __init__(self, model_path, d_model=1024, n_layers=32, input_dim=58, lookback=240):
        # Model initialization and CUDA graph capture
        ...
    def predict(self, features, forecast_days=14):
        # Real-time inference with early-exit confidence filtering
        ...

```

Backtesting, Walk-Forward Validation, and Execution

1. Walk-Forward Validation
   • Rolling Train/Test Splits: Sequentially train and test the model on rolling windows, mimicking real-world deployment.
   • Lag-Aware Backtesting: All features and targets are computed using only information available at each timestamp.
2. Execution and Risk Controls
   • Rule Engine Cross-Check: Neural predictions must satisfy execution gates (e.g., spread friction, chaos override).
   • Fuzzy Logic Sizing: Final position size is computed as a weighted sum of neural and fuzzy logic confidence, scaled by risk constraints.
   • Portfolio-Level Risk: Enforce delta, gamma, vega, and drawdown limits at both trade and portfolio levels.
3. Performance Metrics
   • Sharpe Ratio: Risk-adjusted return.
   • Max Drawdown: Largest peak-to-trough loss.
   • Net Profit-to-Drawdown (NP/DD): Key institutional metric.
   • Turnover: Average absolute change in position size.

Conclusion
By rigorously integrating robust pricing–hedging duality, path wise super hedging, and martingale measure theory with the DeepMamba2 neural architecture, the CondorBrain system achieves a new standard in institutional-grade SPY iron condor trading. The fusion of prediction sets, regime-aware martingale measures, and fuzzy logic-based position sizing ensures that the system is not only robust to model uncertainty and regime shifts, but also interpretable, risk-aligned, and production-ready for real-time M1 inference.
The mathematical derivations, computational physics analogies, and Python implementation templates provided herein offer a blueprint for deploying robust, high-performance iron condor trading systems that maximize Sharpe ratio, minimize drawdown, and deliver superior NP/DD ratios under real-world market conditions.

Key Takeaways:
• Robust Pricing–Hedging Duality: Provides a pathwise, model-agnostic foundation for pricing and hedging under uncertainty, leveraging prediction sets and martingale measures.
• DeepMamba2 Integration: Enables efficient, long-range sequence modeling, regime-aware risk controls, and interpretable multi-head outputs.
• Fuzzy Logic Sizing: Aggregates multi-factor indicator and neural signals into risk-aligned, interpretable position sizes.
• Computational Physics and Topology: Manifold geometry and persistent homology features

enhance regime detection and volatility forecasting.
• Production Engineering: BF16/TF32 mixed precision, CUDA graphs, and memory optimizations enable sub-second inference on institutional datasets.
• Backtesting and Validation: Walk-forward, lag-aware validation ensures robustness and real-world performance.

This synthesis establishes a mathematically rigorous, computationally efficient, and institutionally robust framework for SPY iron condor trading, suitable for both academic scrutiny and production deployment.